



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

August 28, 2023

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 20391
(THIRD REVISION)

EXPIRATION DATE: 2027-07-31

1. GRANTEE: Hexagon Purus LLC
Lincoln, NE
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, mark, sale, and use of a non-DOT specification fully wrapped fiber reinforced composite gas cylinder with a non-load sharing plastic liner that meets the ISO 11119-3 standard except for the design water capacity and working pressure. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce. The safety analyses did not consider the hazards and risks associated with consumer use, use as a component of a transport vehicle or other device, or other uses not associated with transportation in commerce.
 - c. In accordance with 49 CFR 107.107(a), party status may not be granted to a manufacturing permit. These packagings may be used in accordance with 49 CFR 173.22a.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171- 180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 173.301(f) in that a fire protection system is used in lieu of a pressure relief valve and § 173.302(a) in that the use of a non-DOT specification cylinder is not authorized, except as specified herein.
5. BASIS: This special permit is based on the application of Hexagon dated June 21, 2023, submitted in accordance with § 107.123.

Tracking Number: 2023064952

6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Argon, <i>compressed</i>	2.2	UN1006	N/A
Deuterium, <i>compressed</i>	2.1	UN1957	N/A
Helium, <i>compressed</i>	2.2	UN1046	N/A
Hydrogen, <i>compressed</i>	2.1	UN1049	N/A
Neon, <i>compressed</i>	2.2	UN1065	N/A
Nitrogen, <i>compressed</i>	2.2	UN1066	N/A
Methane, <i>compressed or Natural gas, compressed (with high methane content)</i>	2.1	UN1971	N/A

Note: See paragraph 7.c for cylinder service limitations.

7. SAFETY CONTROL MEASURES:

a. PACKAGING: Packaging prescribed is a non-DOT specification fully wrapped fiber reinforced composite gas cylinder with a non-metallic and non-load sharing plastic liner as described in the application on file with the Office of Hazardous Materials Safety (OHMS). Each cylinder must meet all the design and construction requirements for UN composite cylinders specified in § 178.71(l) and in ISO Standard 11119-3 (Gas Cylinders of Composite Construction-Specification and Test Methods – Part 3: Fully wrapped fiber reinforced composite gas cylinders with non-metallic and non-load-sharing metallic liners), except as follows:

(1) Scope § 1: Cylinders manufactured under this special permit are limited:

(i) Minimum water volume of 150 liters and maximum water volume of 450 liters with a working pressure up to 950 bar (13,775 psi); or

(ii) Minimum water volume of 150 liters and maximum water volume of 600 liters with a working pressure up to 555 bar (8,048 psi). See

operational controls for maximum internal pressure at 55 °C (130 °F).

- (2) Batch of non-metallic liners § 3.4: A batch of non-metallic liners is the quantity of liners of the same nominal diameter, length, thickness and design, made successively from the same materials, and subjected to the same manufacturing process.
- (3) Design (General Requirement) § 7.1.4: The minimum fiber stress ratio for carbon fiber must be 2.40.
- (4) Type approval procedure (General Requirement) § 8.1:
 - (i) A DOT Independent Inspection Agency (IIA) approved in writing by the Associate Administrator for Hazardous Materials Safety (AAHMS) in accordance with 49 CFR Part 107, Subpart I must review the results of design qualification testing that was submitted in the application for special permit. The IIA must either verify that the cylinder design meets the requirements of the special permit based on the testing and other documentation submitted in the application for special permit or the IIA may require additional testing and/or information from the manufacturer to verify the cylinder design meets all requirements of the special permit. Prior to production of cylinders, the IIA's verification of the cylinder design must be submitted to and acknowledged in writing by the OHMS.
 - (ii) Prior to any manufacture of cylinders under this special permit, an IIA approved in writing by the AAHMS must provide inspections and verifications of all batch testing and all new design qualification testing in accordance with the requirements of this special permit.
- (5) Prototype tests § 8.2.1, § 8.2.2: The cylinders required to be manufactured for prototype testing must be representative of production units. A sufficient number of tubes shall be made available to complete the prototype testing or testing of the design variant. Subscale units are permitted as follows:
 - (i) Ambient cycling: One test unit must be full scale, additional unit must be full diameter, shorter length;
 - (ii) LBB: test unit must be full diameter and may be shorter length;
 - (iii) Environmental fluid: test unit may be smaller diameter and shorter length;
 - (iv) Environmental cycle: test unit may be smaller diameter and shorter length;

- (v) High temperature creep and accelerated stress rupture: test unit may be smaller diameter and shorter length;
 - (vi) Flaw: test unit must be full diameter and may be shorter length;
 - (vii) Gunfire: test unit must be full diameter and may be shorter length;
 - (viii) Permeability: test unit may be smaller diameter and shorter length;
 - (ix) Torque: test unit must be full diameter and may be shorter length.
- (6) Inspector § 8.2.7: The IIA must witness all testing as specified in this special permit (see Table 2).
- (7) Drop test § 8.2.7h, § 8.5.9: Cylinders manufactured under this special permit are not authorized for shipment unless mounted in a frame and must be handled in accordance with the operational controls listed in this special permit; therefore, they are exempt from drop test requirement of § 8.2.7h.
- (8) Salt water immersion test § 8.2.7m, § 8.5.14: Cylinders manufactured under this special permit are not authorized for underwater use, therefore cylinders are exempt from salt water immersion testing.
- (9) New design § 8.3.2(d): a minor change to a resin component that is within the same specification (i.e., from one epoxy to another) may be qualified as a design variant.
- (10) Design Variant § 8.4: Attached Table 2 (qualification tests) may be used in lieu of Table 2 of ISO 11119-3.

Table 2. Qualification tests for cylinders with maximum test pressure ≤ 1505 bars, water volume greater than 150 liters and less than or equal to 600 liters.

Qualification for Design															Pressure Relief Device
Test	New Design	Length $\leq 50\%$	Length $> 50\%$	Diameter $\leq 20\%$	Diameter $> 20\%$	Liner thickness $> 20\%$ or $\leq 20\%$	Liner material	Equivalent fiber	Test Pressure $\leq 20\%$	Test pressure $> 20\%$	Composite	Boss-to-liner	Equivalent resin matrix	Resin Matrix	
Liner material test							X								
Composite material test								X			X ¹		X ₁		
Hydraulic pressure	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Hydraulic burst	X	X ¹	X ¹	X ¹	X	X ¹	X ¹	X	X ¹	X	X	X ¹	X ¹	X ₁	
Ambient cycle	X			X ¹	X		X	X ₁	X ₁	X	X	X ₁			
Environmental cycle	X													X	
High temperature creep ¹	X				X ₁		X			X ^{1a}	X ^{1a}	X _a	X _a	X _a	

Qualification for Design														Pressure Relief Device
Test	New Design	Length <= 50%	Length >50%	Diameter <=20%	Diameter >20% <=	Liner thickness >20% or	Liner material	Equivalent fiber	Test Pressure <=20%	Test pressure >20%	Composite	Boss-to-liner	Equivalent resin matrix	Resin Matrix
Flaw tolerance 1	X				X									X
High velocity impact (gunfire)	X				X ₂				X ₁	X ₁	X ₁			X
Fire resistance	X		X _{1b}		X _{1b}					X				X
Permeability	X				X ₂ ;	X	X			X ₁				
Torque	X						X					X		X
Leak	X			X	X		X		X	X		X		
Pneumatic cycle	X				X ₁	1	X					X		

Notes:

- For a new design of a cylinder with water volume larger than 450 L, a minimum of 1 cylinder may be used for each design change. For a change of boss-liner interface column, a leak check of the liner interface would be accepted. The pneumatic cycle test is not required if the boss-liner interface does not change.

- a. Where the design variant's burst pressure to test pressure ratio, is over 20% greater than the same ratio for the approved design.
 - b. When length increases up to 50% and/or diameter increases up to 20%, Bonfire test may not be required if the volume stays the same or decreases and the same PRD system is used.
2. Test to be conducted for reduction in diameter only.
- (11) Cylinder burst test § 8.5.3.1: Two cylinders are required for the burst test. Pressurization rate is limited to 14 bar/sec at pressures above 80% of the minimum burst pressure. If the rate exceeds 3.5 bar/sec above 80% of the minimum burst pressure, then either the cylinder must be placed schematically between the pressure source and the pressure measurement device, or there must be a 5 second hold at the minimum design burst pressure.
 - (12) Burst test criteria § 8.5.3.2: The burst pressure or pressure at failure, P_b , must be not less than 1.6 times the test pressure, P_h , of the composite cylinder.
 - (13) Ambient cycle test procedure § 8.5.4.1.1: The cylinders must be cycled to 130% of the working pressure or higher. At least one of these cylinders must be a full-scale cylinder. One of these cylinders may be a subscale with full diameter, but shorter length. The cylinder must pass the LBB test, 8.5.19, with no leakage. Temperature monitoring of cylinder not required if temperature is maintained below 85 °C.
 - (14) Ambient cycle test criteria § 8.5.4.1.2: The cylinder shall withstand a number of cycles equal to 750 times the design service life in years without leakage. The cylinder shall continue cycling until it leaks or reaches a total of 2250 times the service life in years. However, should failure during this second part of the test be by burst, then the cylinder shall have failed the test.
 - (15) Environmental cycle test procedure § 8.5.6.2: The cylinder shall have an internal pressure of 10% of the working pressure during the temperature conditioning. The cylinder shall be pressure cycled between 10% and 100% of the working pressure during the extreme temperature exposure. Following the cycle testing at designated temperature (60-70 °C), the cylinder shall be subjected to 30 cycles from 10% of working pressure to the test pressure at a rate not to exceed 10 cycles per minute.
 - (16) Environmental cycle test criteria § 8.5.6.3: The burst pressure, p_b , must be not less than 1.4 times the test pressure, p_h .

(17) High temperature creep test procedure § 8.5.7.1, 8.5.7.2: One cylinder shall be hydrostatically pressurized to test pressure at 65 °C. The cylinder shall be held at this pressure and temperature for 1,000 hours. The cylinder shall then be pressurized to burst in accordance with the procedure described in 8.5.3, except that the burst pressure shall exceed 100% of the minimum design burst pressure (1.6 times the test pressure, P_h). A second cylinder shall be pressurized to test pressure and held at a temperature of 100 °C for not less than 200 hours. Following the test, the cylinder shall meet the requirements of the hydrostatic expansion test (8.5.2) and the burst test (8.5.3), with the minimum burst being 1.6 times the test pressure, P_h , of the composite cylinder.

(18) Flaw test criteria § 8.5.8.1: One cylinder must be tested. A subscale tank with full diameter and shorter length may be used for the test.

(19) Flaw test criteria § 8.5.8.3: The cylinder must not leak or rupture within the first 3,000 cycles but may fail by leakage during the remainder of the cycles.

(20) Drop test § 8.5.9: The drop test is not required. (See operational controls).

(21) High velocity impact test procedure § 8.5.10.1: The cylinder may be filled with air, nitrogen, or the gas to be contained. If a single 7.62 mm (30 caliber) round will not penetrate the wall, additional rounds may be used, or a larger diameter round may be used.

(22) Fire resistance test § 8.5.11.2: The cylinder (tube) assembly must be tested in a horizontal position. The bonfire test is designed to demonstrate that finished tubes, complete with the fire protection system (e.g., valve, pressure relief devices and/or integral thermal insulation) specified in the design, will prevent the rupture of the tube when tested under the specified fire conditions. A uniform fire source of a minimum 1.65 m length must provide direct flame impingement on the cylinder surface. Any failure during the test of a valve, fitting or tubing that is not part of the intended protection system for the design shall invalidate the result. Direct flame impingement is not permitted on valves, fittings, and/or pressure relief devices. The fire and its position relative to the transport frame must be approved by the Inspector. Precautions shall be taken during fire testing in the event that tube rupture occurs.

(23) Fire resistance test criteria § 8.5.11.3: The cylinder must vent through a pressure relief device, and without rupturing.

(24) Permeability test procedure § 8.5.12.2: One cylinder must be tested. This may be the full-scale cylinder or a subscale (shorter length with same diameter) using the same liner material. Pressure cycling prior to permeation testing is not required. The cylinder must be filled with the intended lading or suitable trace gas

to working pressure, placed in an enclosed sealed chamber at ambient temperature, and monitored for leakage for 500 hours.

(25) Permeability test criteria § 8.5.12.3: The measured permeation rate must be reported. If a trace gas is used, correlation between the trace gas and intended lading must be provided. If permeation exceeds the allowable rate of § 8.5.12.3, the cylinder must be sectioned, and the internal surfaces inspected for any evidence of cracking or deterioration.

(26) Torque test on cylinder neck boss § 8.5.13.1: The body of the cylinder must be restrained against rotation and a torque of twice the valve or PRD installation torque specified by the manufacturer must be applied to each end boss of the cylinder. The torque must be applied first in the direction of tightening a threaded connection, then in the un-tightening direction, and finally again in the tightening direction.

(27) Torque test criteria § 8.5.13.2: Upon completion of all applied torques, the cylinder must be subjected to a leak test to confirm that no leakage occurred as a result of applied torque.

(28) Salt water immersion test § 8.5.14: The salt water immersion test is not required, as the cylinder is not authorized for underwater applications.

(29) Leak test § 8.5.15: An acceptable procedure for leak testing is to pressurize the cylinders with suitable fluid to working pressure and, with the cylinder having been pressurized for at least 15 minutes, carefully examine for signs of leakage (e.g., a visual indication or decrease in pressure).

(30) Pneumatic cycle test procedure § 8.5.16.1: One finished cylinder shall be cycle tested as follows. Temperature monitoring of the cylinder is not required if the temperature is maintained below 85 °C:

(i) Fill the cylinder to be tested with a non-corrosive fluid such as oil, inhibited water, or glycol;

(ii) Cycle the pressure in the cylinder for 1,000 cycles, between 10% of working pressure and working pressure. The pressure cycling rate shall not exceed 10 cycles per minute;

(iii) Release the pressure, drain the fluid, and dry the interior of the cylinder;

(iv) Cycle the pressure in the cylinder for 5 cycles, between 10% of working pressure and working pressure, with air, nitrogen or other gas

determined by the inspector. The pressure cycling rate must provide at least a 2 hour hold at the high-pressure portion of the cycle;

(v) Following the high pressure hold of the final cycle, the gas shall be released freely to atmosphere.

The cylinder must then be subjected to a leak test. Following the completion of the test, the cylinder must be sectioned and the liner and liner/end boss interface inspected for evidence of any deterioration, such as fatigue cracking or electrostatic discharge.

(31) Pneumatic cycle test criteria § 8.5.16.2: The cylinder must have no signs of leakage or deterioration.

(32) Water boil test § 8.5.17: The water boil test is not required.

(33) Batch inspection and testing sampling § 9.1.3: Supplier's certification of the material properties may serve as verification of compliance.

(34) Batch inspection and testing criteria § 9.1.4: Supplier's certification of the liner boss properties may serve as verification of compliance.

(35) Overwrap materials § 9.3: Supplier's certification of the fiber properties may serve as verification of compliance.

(36) Batch testing procedure § 9.4.5: One cylinder (tube) out of 5 batches or 6 months of cylinder production, whichever comes first. A batch here is defined to be production quantity of up to 200 finished cylinders (tubes) successively produced (plus finished cylinders required for destructive testing), of the same nominal diameter, length, thickness and design. The batch of finished tubes may contain different batches of liners, fibers and matrix materials.

(37) Batch testing criteria § 9.4.6: The burst test may be conducted on the first unit of the batch. After reaching the minimum required burst pressure, and holding for 5 seconds, the cylinder shall have passed the test.

(38) Marking § 10.1 & 10.2: marking must contain the following:

(i) DOT special permit number followed by working pressure expressed in bar (psig). Marking may be on a label permanently attached to the outside of the cylinder.

(ii) A serial number and the manufacturer's identification number or a symbol as obtained from the Associate Administrator for Hazardous

Materials Safety, located just below or immediately following the DOT marking. The serial number and the manufacturer's identification number may be placed on the boss provided the markings are accessible for inspection.

(iii) The DOT Independent Inspector Agency (IIA) official mark must be placed near the serial number. The marking must contain date the (month and year) of the initial hydraulic proof pressure test for that cylinder.

(iv) The size of the letters and numbers used must be at least 0.64 cm (1/4 inch) high if space permits.

(v) The following are examples of an authorized format for marking:

DOT-SP AAAAA-YYYY

(where AAAAA is the special permit number and YYYY is the working pressure)

CCCC MMI

(where CCCC is the serial number and MMI is the manufacturer's mark or symbol)

DDD - MM/YY

(where DDD is the inspectors mark and MM/YY is the month and year of the hydraulic proof pressure test).

Additional markings are permitted, provided the additional markings do not obscure the required marking and are not detrimental to the integrity of the cylinder. Provisions for marking of the required requalification dates and RIN information must be made near the cylinder markings.

(39) Additional requirements for each new design:

(i) Fire Protection System (FPS): Each tube assembly must be equipped with a Fire Protection System (FPS) as described in the application on file with the OHMS. The FPS consists of:

(A) Plastic sensor lines that are energized from a low-pressure reservoir, a pressure release mechanism, and vent lines. There are multiple sensor lines, and they run the length of the frame system. When the sensor lines are exposed to a fire, they melt or rupture, causing the air pressure inside the reservoir of the FPS system to drop. The pressure drop in the FPS reservoir activates the tube assembly pressure release mechanism, which vents all tubes; or

(B) FPS system that uses a shape memory metal material as a trigger the vent valve during a fire scenario as specified in the Hexagon patent application on file in the OHMS.

(C) The vent lines direct the released gas upwards and outside of the frame system.

(ii) Environmental fluids test: One finished cylinder including coating if applicable must be tested in accordance with procedures described in the application on file with the OHMSAPD.

(iii) Leak Before Burst (LBB) Test: Two finished cylinders, of full-scale diameter but may be shorter length, must be pressure cycled in accordance with the following procedure:

(A) Fill the cylinder to be tested with a non-corrosive fluid such as oil, inhibited water or glycol;

(B) Between 25 bar and 150% of the working pressure at a rate not exceeding 10 cycles per minute;

(C) The number of cycles to failure must be reported, along with the location and description of the failure initiation; and

(D) All cylinders must either fail by leakage or exceed three times the design number of filling cycles.

(iv) Resin Shear Strength: Resin materials must be tested in accordance with procedures described in the application on file with the OHMSAPD.

(v) Ultraviolet (UV) Testing: One finished cylinder including coating if applicable must be UV tested in accordance with procedures described in the application on file with the OHMS.

b. Requalification: Each cylinder must be requalified once every 5 years by using one of the two methods described in this special permit. The facility that performs requalification of these composite cylinders (tubes) must be a RIN holder for requalification of this type of composite cylinder (tubes) as described in § 180.205(b) or have a valid special permit for requalification of this type of composite cylinders/tubes using Modal Acoustic Emission (MAE) testing.

(1) Method 1: Hydraulic Proof Pressure Testing and Visual Inspections. The requalification facility must hold a RIN for requalification performing hydraulic proof pressure testing and visual inspections and meet the following requirements:

- (i) Knowledge, documentation, equipment and instrumentation for performing the external and internal visual inspection of cylinders manufactured in accordance with the provisions of DOT-SP 20391;
 - (ii) Knowledge, documentation and equipment for performing the proof-pressure testing of cylinders manufactured in accordance with the provisions of DOT-SP 20391;
 - (iii) Adequate facilities, handling equipment, and skills to ensure cylinders manufactured in accordance with the provisions of DOT-SP 20391 will not be subject to impact or other damage during disassembling and reassembling.
 - (iv) Acknowledgment that the requalifier understands the specific operational controls of DOT-SP 20391, paragraph 7.c.(4) that states “cylinder must be rejected if it dropped or impacted during requalification.”
 - (v) Availability to document that during the requalification process, the structural integrity of frame design is not compromised and remains the equal to or greater than the requirements specified in DOT-SP 20391, paragraph 7.c.(3).
 - (vi) Visual Inspections: The external and internal visual inspection must be in accordance with CGA pamphlet C-6.2 and with Hexagon Service Bulletin 10-01-002, Hexagon CNG Bulk Hauling TITAN® Module Inspection Manual on file with the OHMS.
 - (vii) Hydraulic proof pressure test as described in CGA Pamphlet C.1, which the test pressure is equal to 1.5 times the marked working pressure and hold the pressure for a minimum of 3 minutes without a loss of pressure. The testing facility for proof pressure test must be equipped with protection system (e.g., water jacket well or concrete barrier) to avoid injury during requalification process.
- (2) Method 2: Modal Acoustic Emission (MAE) Testing and External Visual Inspection. The requalification facility must hold a RIN for requalification performing MAE and external visual inspection and must meet the following requirements:
- (i) Holder of a special permit in performing MAE testing on composite cylinders (tubes).

(ii) MAE testing must in accordance with the Modal Acoustic Emission (MAE) Examination Procedure for Requalification of Composite Overwrapped Pressure Vessels (Cylinders and tubes) posted on PHMSA website, <https://www.phmsa.dot.gov/technical-resources/hazmat-technical-resources/technical-reports> and the additional procedure provided by Hexagon Bulletin 17-05-001 on file with the OHMS.

(iii) External visual inspection must be in accordance with CGA pamphlet C-6.2 and with Hexagon Service Bulletin 10-01-002, Hexagon CNG Bulk Hauling TITAN® Module Inspection Manual on file with OHMS.

(iv) Tubes involved in a tipover or rollover accident: For tubes that were possibly subjected to impact damage from an event such as tube trailer collision or rollover accident, the pressurization of the MAE testing must be by hydraulic medium (e.g., water) rather than gaseous medium.

(3) Requalification Marking: Date (month/year) must be permanently marked on the cylinder as specified in paragraph § 180.213. The marking of the RIN symbol on the cylinder certifies compliance with all the terms and conditions of this special permit.

c. OPERATIONAL CONTROLS:

(1) Cylinders manufactured under this special permit are not authorized for use 15 years from the date of manufacture, except as specified under paragraph 8.a. of this special permit.

(2) A cylinder that has been subjected to fire may not be returned to service.

(3) Cylinders are permanently mounted:

(i) Inside of framing that is designed, marked (approval plate) and approved in accordance with the International Convention for Safe Containers (CSC) (49 CFR Part 451) as described in the application on file with the OHMS. Structural framework has been evaluated for transportation of the tubes under this special permit by Finite Element Analysis (FEA) on file with OHMS. The FEA has demonstrated the framework's ability to protect the tubes from damage due to front, rear, or side impact, and rollover. The frame designed meets the following:

(A) All requirements of § 173.301(i);

(B) The frame design must meet all requirements of CGA TB-25; or

(ii) In a container or semitrailer that is not approved and marked per CSC that shall be designed and analyzed using Finite Element Analysis (FEA) software. Analysis shall demonstrate the ability of the framework structure to protect the tubes from damage due to front, rear, or side impact, and rollover. Structural framework Finite Element Analysis (FEA) shall be on file with the OHMS. This container or semitrailer design meets the following:

(A) All requirements of § 173.301(i).

(B) The frame design must meet all requirements of CGA TB-25.

(C) Except for the requirements for length, width, height and ISO/CSC markings, the frame is also tested in accordance with ISO 1496-3:1995(E) per the following sections:

1. Section 6.3 - Test No. 2, Top Lift;
2. Section 6.4 - Test No. 3, Bottom Lift;
3. Section 6.5 - Test No. 4, External Restraint (Longitudinal);
4. Section 6.6 - Test No. 5, Internal Restraint (Longitudinal);
5. Section 6.7 - Test No. 6, Internal Restraint (Lateral);
6. Section 6.8 - Test No. 7, Rigidity (Transverse) (for containers 20' and longer);
7. Section 6.9 - Test No. 8, Rigidity (Longitudinal) (for containers 20' and longer);
8. Section 6.11 - Test No. 10, Walkways (if applicable);
9. Section 6.13 - Test No. 12, Pressure Test.

(4) Cylinder (tube) handling: Cylinder must be rejected if it drops from a height greater than 2' during the manufacturing and/or prior to being mounted to the CSC framing.

(5) Fire protection System (FPS) Inspection: The FPS must be inspected annually in accordance with Hexagon Service Bulletin SB 16-04-002–Titan Venting System and Fire Protection System and Maintenance Check manuals. During cold temperature, where freezing may occur, all vent lines must be inspected each week to make sure they are free from debris and the vent caps O-rings are attached to prevent water into to the vent lines. The annual inspection and maintenance of the FPS and its operation must be in accordance with the following product operational manuals on file with OHMSAPD:

- (i) TITAN® IV Normally Open FPS – SB 12-09-001;
- (ii) TITAN® IV Normally Closed FPS – SB 14-02-005;
- (iii) TITAN® IV BAV FPS – SB 15-01-002;
- (iv) TITAN® XL – SB 15-10-001.

All the inspections documents must be available for review to OHMS per request.

(6) Cabinet Flammability Limit: Lower Level Flammability Limit (LEL) of each gas or gas mixtures must be calculated for the highest pressure and temperature and to ensure the cabinet of the cylinder assembly is equipped with proper ventilation to avoid a fire or explosion during transportation.

(7) Prior to use in Offshore Service under the terms of this special permit, additional information justifying such use must be submitted to and acknowledged in writing by the AAHMS.

(8) Low pressure/temperature prior to filling: The following procedure must be followed in case the pressure of a cylinder (tube) dropped below 100 psig (7 bar) while the ambient temperature was below -12 °C:

Prior to filling: either the tube must be held at or above 16 °C for 8 hours, or the tube must be filled to 435 psig (30 +/- 3 bar) from a compressor, and held for one hour, before returning to normal fill procedures.

(9) Transportation of Division 2.1 (flammable gas) is not authorized aboard cargo vessel unless specifically authorized in the Hazardous Materials Table (§ 172.101).

(10) When transported by cargo vessel, the cylinders must be stowed on deck only and are prohibited from passenger ships (Stowage Category D).

(11) Filling procedure for CNG: All cylinders/tubes used for CNG (UN1971) service must meet the procedure and conditions described in Hexagon CNG Bulk Hauling Operation and Inspection Manual, on file with the OHMS and the following:

- (i) Maximum dew point of the gas -9 °C (15.8 °F).
- (ii) Maximum carbon dioxide (CO₂) content 4%.
- (iii) Maximum Oxygen (O₂) content 1%.
- (iv) Maximum temperature during filling 70 °C (158 °F).
- (v) Maximum pressure during transportation $\leq 5/4$ times Working pressure.

(12) Cylinder (tube) that exhibits liner bulge: Liner bulge must be corrected as follows:

- (i) Pressurize the tube to 10% of its marked working (service) pressure and hold for a minimum of 4 hours. Then depressurize the tube, perform an internal visual inspection and ensure no liner bulge is present.
- (ii) If a liner bulge is still present after the first pressurization described above, take the following actions:
 - (A) Pressurize the tube to its marked working (service) pressure and hold for a minimum of 1 hour. Then depressurize the tube, perform an internal visual inspection and ensure no bulge is present in the liner;
 - (B) If a liner bulge is still present after the second pressurization as described above, the tube must be rejected;
 - (C) For a rejected tube, contact the tube manufacturer to obtain additional guidance in correcting the liner bulge prior to marking the tube

8. SPECIAL PROVISIONS:a. Service Life Extension Program:

(1) Cylinders manufactured under this special permit are authorized for a maximum service life of 15 years from the date of manufacture in accordance with the Hexagon service life extension program dated February 14, 2013, on file with the OHMS. The service life extension program must be implemented for each design type that is intended for additional service life beyond 15 years to determine the additional years of service life. If cylinders are authorized for extended service life, the maximum service life of each cylinder under this special permit is 30 years from the date of manufacture.

(2) Under the service life extension program, the grantee must randomly recall a minimum of thirty cylinders of each design type which have been in service for 10 and 13 years. Cylinders recalled after 10 years shall be designated "Group A" and cylinders recalled after 13 years shall be designated "Group B". All recalled cylinders must be subjected to design requalification as specified Sections 8.5.4, 8.5.5, 8.5.7 and 8.5.8 of ISO 11119-3. Acceptance criteria shall be as defined in ISO 11119-3 except $P_b = 1.6P_h$ and the design life (y) must be greater than or equal to 20 years. All cylinders that fail to meet the requalification requirements must be condemned, removed from service and rendered incapable of retaining pressure. In the case that some units from the initial minimum lot size are condemned, an additional 30 cylinders must be selected and subjected to the same design requalification as specified above (Sections 8.5.4, 8.5.5, 8.5.7 and 8.5.8 of ISO 11119-3). An Independent Inspection Agency must witness all testing.

(3) The complete test report including original test data must be submitted to the Associate Administrator for Hazardous Materials Safety for assessment within 30 days of completion. Failure to meet the acceptance criteria specified in this section shall result in the design being restricted to a maximum life of 15 years.

b. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

c. A person who is not a holder of this special permit, but receives a package covered by this special permit, may reoffer it for transportation provided no modification or change is made to the package and it is offered for transportation in conformance with this special permit and the HMR.

- d. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.
 - e. A current copy of this special permit must be maintained at each facility where the package is manufactured under this special permit and must be made available to a DOT representative upon request.
 - f. Each packaging manufactured under the authority of this special permit must be either (1) marked with the name of the manufacturer and location (city and state) of the facility at which it is manufactured or (2) marked with a registration symbol designated for a specific manufacturing facility.
9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, and cargo vessel.
10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each cargo vessel or motor vehicle used to transport packages covered by this special permit. For transportation by cargo vessel, see paragraphs 7.c.(9) and 7.c.(10).
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:
- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
 - o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
 - o Registration required by § 107.601 et seq., when applicable.

Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)— “The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. **REPORTING REQUIREMENTS:** Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Material Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: M Toughiry/Casey Chambers